

Techno-economic analysis of metal-organic frameworks for hydrogen and natural gas storage

Daniel DeSantis,¹ Jarad A. Mason,^{2,3} Brian D. James,¹ Cassidy Houchins,¹ Jeffrey R. Long,^{2,4} and Mike Veenstra⁵

¹Strategic Analysis, Inc., Arlington, VA 22203, USA. ²Department of Chemistry, University of California, Berkeley, CA 94720, USA, ³Materials Sciences Division, Lawrence Berkeley National Laboratory, Berkeley, CA 94720, USA,

⁴Department of Chemical and Biomolecular Engineering, University of California, Berkeley, CA 94720, USA, and ⁵Ford Motor Company, Research and Advanced Engineering, Dearborn, MI 48121, USA.

Metal Organic Framework, MOF, Cost, Natural Gas Storage, Hydrogen Gas Storage, Manufacturing, Liquid Assisted Grinding, LAG, Economic Analysis

ABSTRACT: A techno-economic analysis was conducted for metal-organic framework (MOF) adsorbents, which are promising candidates for light-duty vehicle onboard natural gas and hydrogen storage. The goal of this analysis was to understand cost drivers for large-scale (2.5 Mkg/year) MOF synthesis and to identify potential pathways to achieving a production cost of less than \$10/kg MOF. Four MOFs were analyzed with four different metal centers and three different linkers: Ni₂(dobdc) (dobdc⁴⁻ = 2,5-dioxido-1,4-benzenedicarboxylate; Ni-MOF-74), Mg₂(dobdc) (dobdc⁴⁻ = 2,5-dioxido-1,4-benzenedicarboxylate; Mg-MOF-74), Zn₄O(bdc)₃ (bdc²⁻ = 1,4-benzenedicarboxylate; MOF-5), and Cu₃(btc)₂ (btc³⁻ = 1,3,5-benzenetricarboxylate; HKUST-1). Baseline costs are projected to range from \$35/kg to \$71/kg predicated on organic solvent (solvothermal) syntheses using an engineering scale-up of laboratory demonstrated synthesis procedures and conditions. Two alternative processes were analyzed to evaluate the cost impact of reducing solvent usage: liquid assisted grinding (LAG) and aqueous synthesis. Cost projections¹ from these alternative synthesis approaches range from \$13/kg to \$36/kg (representing 34%-83% reductions), demonstrating the large impact of solvent on the baseline analysis. Finally, sensitivity studies were conducted to identify additional opportunities for achieving MOF production costs of less than \$10/kg.

¹Cost refers to production cost, including materials and manufacturing, but does not include a mark-up.

INTRODUCTION

Hydrogen (H₂) and natural gas (NG) are attractive gasoline alternatives owing to their potential for reducing greenhouse gas emissions, addressing dependence on foreign sources of fuel, and reducing the total cost of vehicle ownership. While H₂ has a high specific energy (energy per unit mass) relative to gasoline and NG has a comparable specific energy to gasoline, both H₂ and NG have much lower energy densities (energy per unit volume) than gasoline. This poses a significant challenge to conventional compressed gas storage. As transportation fuels, H₂ and NG are typically stored at 700 bar and 250 bar nominal working pressure, respectively.¹⁻³ On a volumetric basis, H₂ at 700 bar requires 7 times greater storage volume and NG at 250 bar requires 3.5 times greater storage volume to yield the same energy content as a liter of gasoline. While this gap may be somewhat offset for a fuel cell vehicle with improved powertrain efficiency, increased onboard storage energy density is needed for both H₂ and NG to achieve an equivalent driving range with conventional vehicles. To address these challenges, the Department of Energy (DOE) has set 2020 technical targets for hydrogen storage systems cost (\$333/kg H₂), volumetric capacity (40 g H₂/L), and gravimetric capacity (5.5 wt % H₂).⁴ Similarly, to be competitive with the incumbent technologies, the DOE Advanced Research Projects Agency-Energy (ARPA-E) developed NG storage targets for system cost (\$150/GGE), volumetric capacity (148 g NG/L), and gravimetric capacity (31 wt % natural gas).⁵ Compressed gas storage falls far short of the DOE cost and capacity targets due to the expensive and bulky pressure vessels, so alternative storage technologies are being explored.¹

One strategy to increase the volumetric capacity of alternative fuel vehicles (AFVs) is to bind H₂ and CH₄ to an adsorbent surface through a weak van der Waals interaction (physisorption).⁶ Metal-organic frameworks (MOFs) are one such class of adsorbents being considered. MOFs are high surface area materials containing a network of metal cations or clusters connected through bridging organic ligands. Owing to their high surface area and tunable surface chemistry, MOFs have received considerable attention as a material-based adsorbent option for natural gas and hydrogen AFVs and have shown potential for achieving high volumetric and gravimetric storage capacities.^{7,8} MOFs are able to store appreciable amounts of NG or H₂ at moderately low pressures (less than 100 bar).⁶ Low-pressure adsorbent-based fuel systems could reduce the cost of on-board tanks, the cost and technical difficulty of high-pressure compressors at the station, and the barriers to building a fuel delivery infrastructure.⁹ MOFs are generally considered to have the potential to improve volumetric and gravimetric density by physisorption through achieving a favorable balance of large surface area and pore volume with high-affinity gas adsorption sites.⁶ BET surface areas for various MOFs can exceed 7,000 m²/g, while the pore volume can range from 0.043-4.40 cm³/g.^{10,11} ARPA-E technical targets for CNG vehicles include delivery rates (2.6 kW/L) and adsorbent gravimetric capacity (0.5 g_{NG}/g_{MOF}),⁵ and \$10/kg_{MOF} target for MOF production for use in NG vehicles.¹² Given the similarity in requirements between CNG vehicles and H₂ vehicles, it is expected that H₂ vehicles will need to achieve similar MOF production costs to be competitive. In fact, with the aforementioned DOE targets for H₂, one can determine that DOE's target H₂ storage system cost is approximately \$18/kg, setting an upper bound on MOF material costs.

To date, the majority of research efforts related to MOFs for NG and H₂ storage have been directed toward the development of new materials exhibiting improved adsorption properties with little regard to material or system cost. Indeed, conventional solvothermal MOF syntheses generally involve the reaction of hydrated metal salts with bridging organic ligands in expensive organic solvents at 100-150 °C.¹³ Recently, several alternative processing techniques have been explored to reduce or eliminate organic solvent usage. Liquid Assisted Grinding (LAG) has been proposed, and recently demonstrated, and offers a pathway to reduced MOF costs by lowering synthesis times and dramatically reducing the amount of solvent needed.^{13,14,15} Aqueous synthesis routes for M₂(dobdc), which substitute water for the much higher priced organic solvents typically used, have been recently demonstrated and represent yet another alternative to conventional synthesis.¹⁵

Table 1 – Chemical names and abbreviations for MOF linkers found in this study.

Linker	Linker Abbr.	MOF Name	Alternative MOF Name (M=Metal cation)
2,5-dioxido-1,4-benzenedicarboxylate	dobdc ⁴⁻	M ₂ (dobdc) (M = Mg, Ni)	M-MOF-74
1,4-benzenedicarboxylate	bdc ²⁻	Zn ₄ O(bdc) ₃	MOF-5
1,3,5-benzenetricarboxylate	btc ³⁻	Cu ₃ (btc) ₂	HKUST-1

In this work, detailed cost analyses were performed for four metal-organic frameworks that have promising adsorption properties for NG and/or H₂ storage: M₂(dobdc) (M = Mg, Ni), MOF-5, and HKUST-1.^{6, 10, 16, 17} (See Table 1 for a summary of the linker abbreviations.) Following a brief discussion of the analysis methods used, the results are organized as follows. First, baseline costs are established using an engineering scale-up of laboratory demonstrated solvothermal synthesis methods. Once the baseline is established, alternative synthesis approaches (LAG and aqueous) are analyzed. The purpose of this paper is to demonstrate general MOF cost trends rather than to provide a detailed cost comparison between MOFs. To facilitate a thorough discussion of the cost drivers and potential savings, Mg₂(dobdc) is presented as a representative example. The trends observed for Mg₂(dobdc) also apply to the other three MOFs analyzed, and we expect these general trends to be similar for other MOFs. This work identifies potential pathways to reducing MOF costs to less than \$10/kg and identifies areas where the MOF research community can focus attention to address cost reduction.

Methods

MOF Cost Calculation Methods

A process-based cost estimation methodology was used to assess the production cost of four representative MOFs: Mg₂(dobdc), Ni₂(dobdc), MOF-5, and HKUST-1. This process-based methodology mimics the actual steps of production and determines final cost by summing the individual costs

incurred in each manufacturing step (i.e. synthesis, filtration, and drying) of the process. A brief description of the approach follows.

For each identified step in the production process, a cost is tabulated based on the materials used in that step, the cost of the capital equipment for the step, and the machine and labor operational time to complete each step. Price quotes for each of the chemical reagents and the necessary capital equipment are collected from suppliers. Operational time calculations are determined from detailed mass and energy balances (including reaction kinetics and yield) for the system along with product information on equipment and reaction cycle times, and standard operating procedure (SOP) for chemical operations. Standard labor and manufacturing rates, as well as utility prices, are used to determine the final manufacturing cost of the product.¹⁸ Cost is assessed at a range of MOF annual production rates, from 50,000 kg MOF/year to 2.5 million kg/year. Each projected MOF cost correlates to a production facility optimized for that level of production: MOF costs are not merely the costs from a large facility operated at lower than design capacity. At a nominal level of 50 kg of MOF per vehicle storage system, this correlates to 1,000 systems per year and 500,000 systems per year, respectively.

For standardized materials and devices, price quotations from industry as a function of annual order quantity form the basis for financial estimates. A learning curve formula is then applied to the available data gathered from industry to provide price estimates between the quotation data points.

When non-standard materials and devices are needed, costs are estimated based on detailed DFMA® style models developed for a specific, fully-defined, manufacturing process train. In this approach, the estimated capital cost, C_{Est} , of a manufactured device is quantified as the sum of materials costs C_{Mat} and the manufacturing costs, C_{Man} . Given that in a chemical process, no tooling or assembly is usually required, these costs are not considered in the technical economic analysis. However, a contingency cost was added, which increases the manufacturing cost by 10%:

$$C_{Est} = C_{Mat} + C_{Man} \quad (1)$$

The materials cost is derived from the amount of raw materials needed to make each intermediate or chemical product, based on the system physical design (material, geometry, and manufacturing method). The manufacturing cost is derived from a specific design of a manufacturing process train necessary to make all parts.

Models were iteratively developed with the technical analysis informing the cost analysis. Process designs were created for each system analyzed. Designs were iterative and further assumptions were made to reduce costs, which then further adjusted the process designs. Generic design descriptions for various synthesis methods are provided below.

Analyzed Production Methods

The manufacturing process for the baseline cost assessment for all four MOFs ($\text{Ni}_2(\text{dobdc})$, $\text{Mg}_2(\text{dobdc})$, MOF-5, and HKUST-1) is defined in Figure 1 as the Industrial Baseline Process. This process is based on lab scale synthesis but has been modified to translate the steps to standard operations conducted in large production facilities. Thus, the Industrial

Baseline Process is intended to represent the cost if the proven lab-scale synthesis was transferred directly to scale-appropriate unit operations. The remainder of this section will describe what assumptions were made to scale up lab demonstrated synthesis. Following the description of the lab to industrial scale up, the rationale and assumptions used for aqueous synthesis and liquid assisted grinding will be described.

Laboratory Methods

A generalized laboratory scale solvothermal synthesis is based on previously reported syntheses,^{17,19} as described here briefly:

1. The metal salt and linker are separately dissolved in organic solvents at less than 5 wt % solid to solvent ratio.
2. The two solutions are combined in a precipitation reaction to form the MOF particles, typically at elevated temperature ($80\text{-}120^\circ\text{C}$)² and 1 atm pressure for 18-20 h.
3. The products are separated from the mother liquid by centrifugation.
4. Products are washed multiple times in DMF and then methanol (20 mL volumes each time).
5. The product MOF is dried in several stages over the course of 31 h.

² Tranchemontagne et al. have demonstrated a successful room temperature synthesis. Further, HKUST-1 was modeled at 25°C , as described in the laboratory methods.

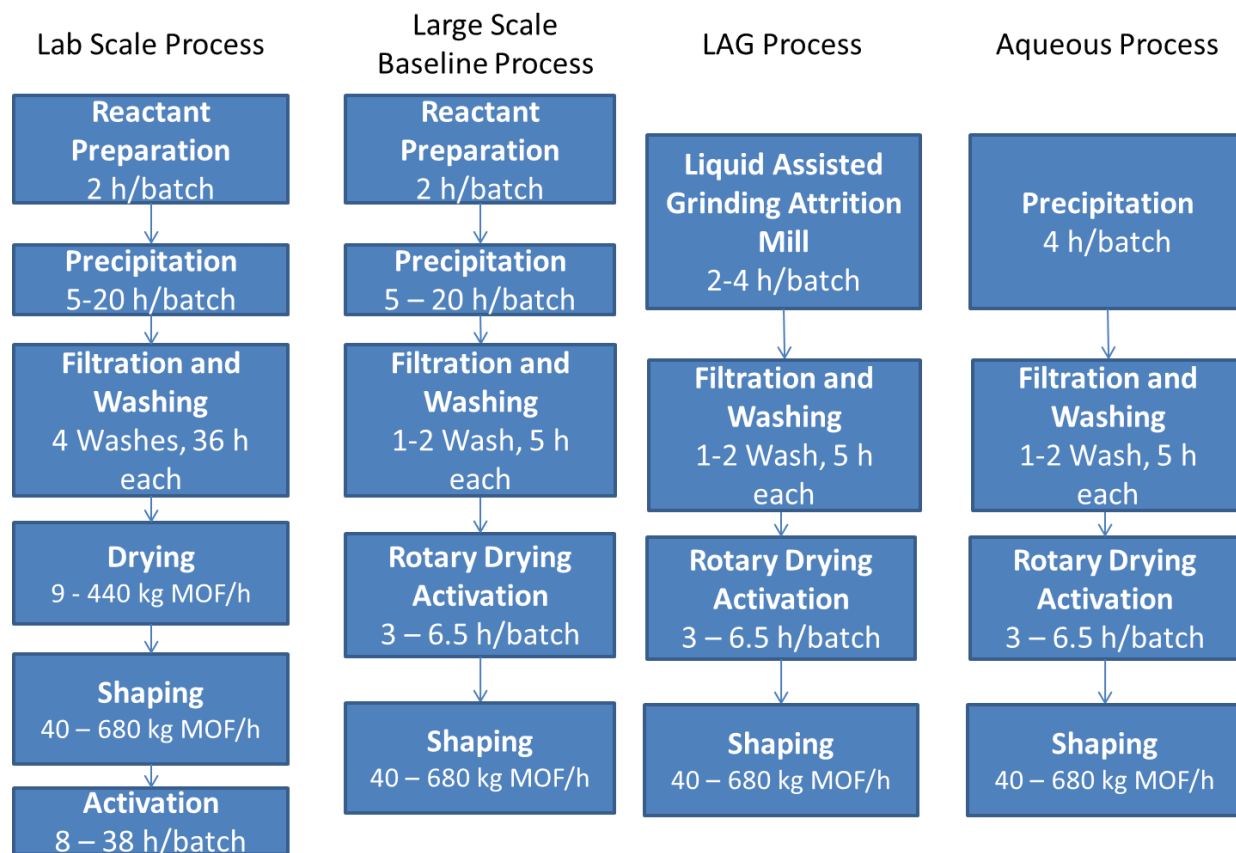


Figure 1 - Block Flow Diagrams for each of the MOF synthesis processes. The processes described cover a range of options to produce MOFs. In this paper, specific MOFs were linked with specific manufacturing processes (See Table 1). Times listed in each step are ranged across the four MOFs analyzed.

The cost analysis for MOF synthesis at 2.5 Mkg/year that follows is based on assumed reaction conditions and process steps demonstrated at lab scale, as described above. However, laboratory procedures are not well suited to high production rates, so three variations on the laboratory procedures were also examined: industrial scale MOF synthesis in organic solvents, synthesis in water, and synthesis using liquid assisted grinding (LAG). Figure 1 summarizes the approaches for each of these four processes with justification for departures from the lab scale synthesis described below.

Industrial Scale Solvothermal Methods

The Industrial Baseline process is based on reaction conditions and the process steps described in the previous section with minor modifications based on engineering judgment. At the industrial scale, it is assumed that the drying and vacuum activation steps can be combined using a rotary dryer. Additionally, a rotary dryer would provide the capacity to combine drying and activation into one step and obviate the liquid-MOF mixing step required for a spray dryer. Furthermore, the rotary dryer has the potential to decompose and/or sublime excess organic ligands that might remain in the framework pores after the filtration/wash step. This contaminant cleansing effect is expected to be more effective than that achievable in a spray dryer or belt dryer, due to the much longer residence time at temperature. A comparison of generalized lab scale solvothermal reaction conditions and our

assumed industrial scale solvothermal reaction conditions is summarized in Table 2. Metal precursors were generally selected for scale up from demonstrated lab-scale solvothermal syntheses. One departure was for $\text{Mg}_2(\text{dobdc})$ and $\text{Ni}_2(\text{dobdc})$. In a typical solvothermal synthesis, $\text{Mg}_2(\text{dobdc})$ and $\text{Ni}_2(\text{dobdc})$ are prepared from $\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ or $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, respectively; however, we assumed that the same synthesis demonstrated by Tranchemontagne et al. and coworkers for $\text{Zn}_2(\text{dobdc})$ using Zn acetate is valid for $\text{Mg}_2(\text{dobdc})$ and $\text{Ni}_2(\text{dobdc})$ using magnesium nitrate ($\text{Mg}(\text{NO}_3)_2$) and nickel nitrate ($\text{Ni}(\text{NO}_3)_2$) salts, respectively.¹⁷ This is expected to be a valid assumption because of the similarities between these metal salts. Caskey et al. reported reaction temperatures of 120 °C for $\text{M}_2(\text{dobdc})$.¹⁹

For solvothermal synthesis, the linker and metal salt are separately dissolved in organic solvents, and the two solutions are combined in a precipitation reactor. Reaction times range from 5 h (MOF-5) to 24 h (HKUST-1), depending on the individual MOF being synthesized. Solid precipitation of the MOF occurs with the solids fraction estimated to be approximately 2.5% by weight of the solution. The solids are filtered and washed. Multiple washes are used to clean excess reagents from the solids followed by a rotary dryer to completely dry the solid MOF. The dry MOF powder can then be compacted into a usable shape (e.g. pellets) and undergo a heated activation step. Once activated, the MOF is ready for use in the onboard NG or H_2 storage system.

Table 2- Laboratory vs. Industrial Baseline Synthesis Process Conditions. Full process conditions for all four analyzed MOFs can be found in supplemental information

	Unit	Laboratory Values	Mg ₂ (dobdc) Industrial Baseline Synthesis Assumptions
Molar Yield	%	69	69
Linker: Metal Salt molar ratio	-	0.5 : 1	0.5 : 1
Precipitation Reactor Pressure	bar	1	1
Precipitation Reactor Temperature	°C	25 to 120	120
Precipitation Reaction Time	h	18	20
Washing Steps	-	5	2
Wash Fluid		DMF/ Methanol	DMF
Drying Time	h	31	6.5

Metal precursors were generally selected for scale up from demonstrated lab-scale solvothermal syntheses. One departure was for Mg₂(dobdc) and Ni₂(dobdc). In a typical solvothermal synthesis, Mg₂(dobdc) and Ni₂(dobdc) are prepared from Mg(NO₃)₂·6H₂O or Ni(NO₃)₂·6H₂O, respectively; however, we assumed that the same synthesis demonstrated by Tranchemontagne et al. and coworkers for Zn₂(dobdc) using Zn acetate is valid for Mg₂(dobdc) and Ni₂(dobdc) using magnesium nitrate (Mg(NO₃)₂) and nickel nitrate (Ni(NO₃)₂) salts, respectively.¹⁷ This is expected to be a valid assumption because of the similarities between these metal salts. Caskey et al. reported reaction temperatures of 120 °C for M₂(dobdc).¹⁹

Liquid Assisted Grinding (LAG) Method

Liquid Assisted Grinding (LAG) is a high volume chemical synthesis technique for preparing co-precipitates in a minimum of solvent that has been demonstrated for preparing MOFs, including HKUST-1.¹⁴ The role of the solvent depends on the particular reaction: in some cases, it acts as a lubricant and energy transfer medium, while in others it acts as a catalyst. Selection of appropriate solvent ratios and solvent mixtures is an active area of research for LAG synthesis.^{20,21,22} Based on advice from experts in LAG manufacturing,²⁰ we assume that the solvent is a 1:1 mass mixture of DMF and methylamine, and that 7.5 wt % (solvent to MOF) is sufficient to wet the solid reactants when added together with the MOF precursor salts in a high energy attrition mill. The mill imparts mechanical energy sufficient to overcome the activation energy of the chemical reaction.^{23–25} LAG may also be a useful means of tuning or controlling surface area and pore volume;²⁴ however, this analysis assumes that the MOF surfaces remain unchanged by the preparation method. The LAG method was modeled only for MOF synthesis, not for linker synthesis.

In the following LAG analysis, ball milling units replaced the precipitation reactor used in solvothermal methods, while the remaining steps of the solvothermal process remained the same. Likewise, reported solvothermal yields (See Table 1) were maintained for LAG calculations.

Aqueous Solution Chemistry

Aqueous synthesis routes have been demonstrated for Ni₂(dobdc) and Mg₂(dobdc) with yields above 90%.^{15,26} Aqueous synthesis is modeled as one pot synthesis with the metal salt and linker dissolved together in deionized water in a precipitation reactor and held at reflux. The filtration, washing, drying, and activation steps are similar to solvothermal synthesis.

Linker Synthesis

Price quotes were obtained for the linkers for MOF-5 (H₂bdc) and HKUST-1 (H₃btc). However, the linker for M₂(dobdc), (H₄dobdc), is not produced commercially on an industrial scale. Consequently, a process based cost estimation analysis was conducted to estimate the cost of H₄dobdc, based on scale-up of a patent production process.²⁷ The modeled manufacturing process involves using a precipitation reactor to produce the linker, which is then filtered and washed before running the powder through a rotary dryer. Reactants, material usages, reaction times, and yields were based on the experimental values described in the patent. Price quotes or estimates were obtained for the raw materials as well as all required capital equipment. The total cost of production of H₄dobdc, as well as the raw materials costs for H₄dobdc, can be found in the supplemental information along with other MOF raw material costs.

Results

Cost results comparing four MOFs (Ni₂(dobdc), Mg₂(dobdc), MOF-5 and HKUST-1) manufactured by the baseline solvothermal synthesis are presented in Figure 2. The cost to manufacture each MOF was calculated based on the respective materials, process conditions, and times presented in Table 2. Cost results presented in Figure 2 are based on the solvothermal synthesis and the Industrial Baseline process diagrammed in Figure 1. Material costs reflect the costs of raw materials (salts, linkers, and solvents), while manufacturing costs reflect the cost of machinery amortized over equipment lifetime as well as process energy, utilities, labor, and facility costs. Solvent costs dominate the material costs for solvothermal synthesis for almost all of the MOFs listed. HKUST-1 is the single exception due to the exceptionally high linker price.³ Material costs also contribute more to the total production cost than manufacturing costs for all cases. (Material cost details can be found in Supplemental Information Table 8.)

³ There is the potential to lower the linker price by manufacturing an appropriate HKUST-1 linker rather than purchasing a commercial linker. This would offer potential cost reductions through the advantage of economies of scale as well as removing a layer of cost mark-up.

Table 3 - Synthesis parameters for the MOF synthesis pathways analyzed

MOF Identity	Linker	Linker Solvent	Metal Salt	MOF Solvent	Synthesis Method	Yield
MOF-5 (Zn ₄ O(bdc) ₃)	H ₂ bdc	DMF ⁴	Zn(O ₂ CCH ₃) ₂ ·2(H ₂ O)	DMF	Solvothermal	63%
HKUST-1 (Cu ₃ (btc) ₂)	H ₃ btc	DMF, EtOH, H ₂ O	Cu(NO ₃) ₂ ·2.5H ₂ O	DMF, EtOH, H ₂ O	Solvothermal	44%
Ni ₂ (dobdc)	H ₄ dobdc	DMF	Ni(NO ₃) ₂ ·6H ₂ O	DMF, MeOH	Solvothermal	69%
Mg ₂ (dobdc)	H ₄ dobdc	DMF	Mg(NO ₃) ₂ ·6H ₂ O	DMF, MeOH	Solvothermal	69%
MOF-5 (Zn ₄ O(bdc) ₃)	H ₂ bdc	DMF, CH ₃ NH ₂	Zn(O ₂ CCH ₃) ₂ ·2H ₂ O	DMF, CH ₃ NH ₂	LAG	63%
Ni ₂ (dobdc)	H ₄ dobdc	DMF, CH ₃ NH ₂	Ni(NO ₃) ₂ ·6H ₂ O	DMF, CH ₃ NH ₂	LAG	69%
Mg ₂ (dobdc)	H ₄ dobdc	DMF	Mg(NO ₃) ₂ ·6H ₂ O	DMF	LAG	69%
HKUST-1 (Cu ₃ (btc) ₂)	H ₃ btc	DMF, EtOH, H ₂ O	Cu(NO ₃) ₂ ·2.5H ₂ O	DMF, EtOH, H ₂ O	LAG	44%
Ni ₂ (dobdc)	H ₄ dobdc	H ₂ O	Ni(CH ₃ COO) ₂ ·4H ₂ O	H ₂ O	Aqueous	92%
Mg ₂ (dobdc)	H ₄ dobdc	H ₂ O	Mg(NO ₃) ₂ ·6H ₂ O	H ₂ O	Aqueous	92%

⁴ Dimethylformamide

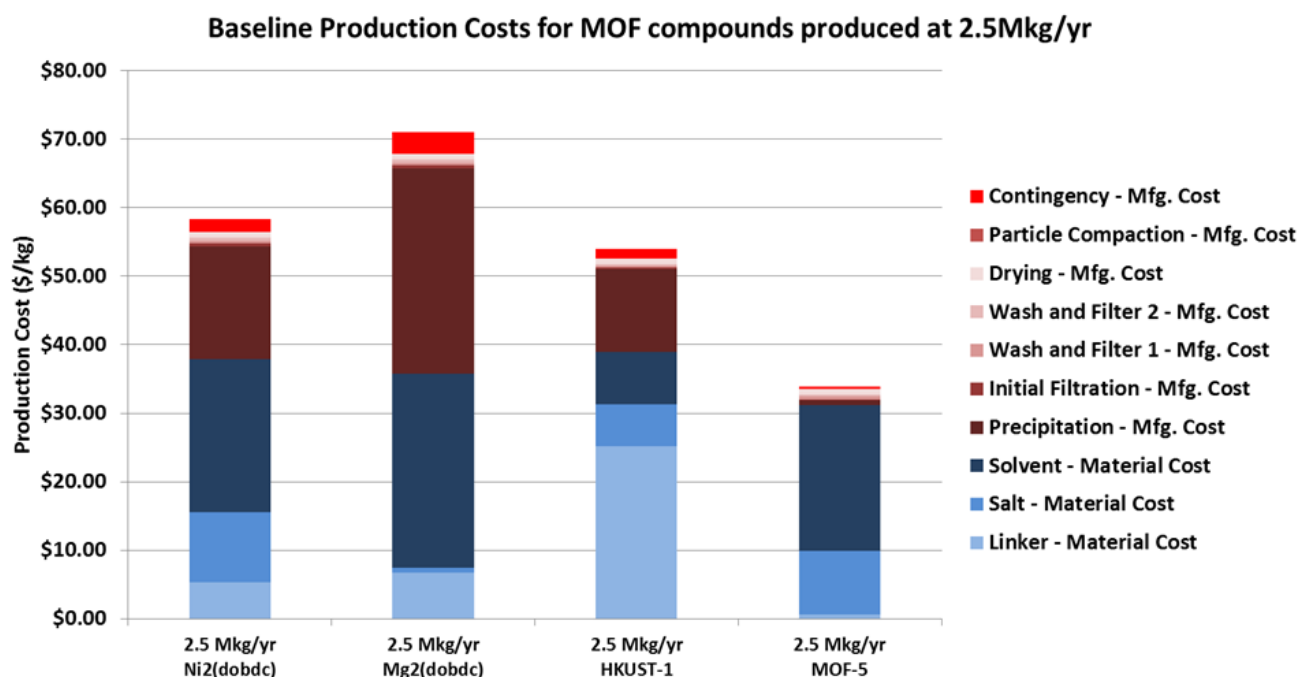


Figure 2 - Baseline total cost of four MOFs manufactured at 2.5 Million kg/year by solvothermal synthesis methods. Material cost contributions (linker, metal salts, and solvents) are shown as shades of blue while manufacturing costs for each step are shown as shades of red.

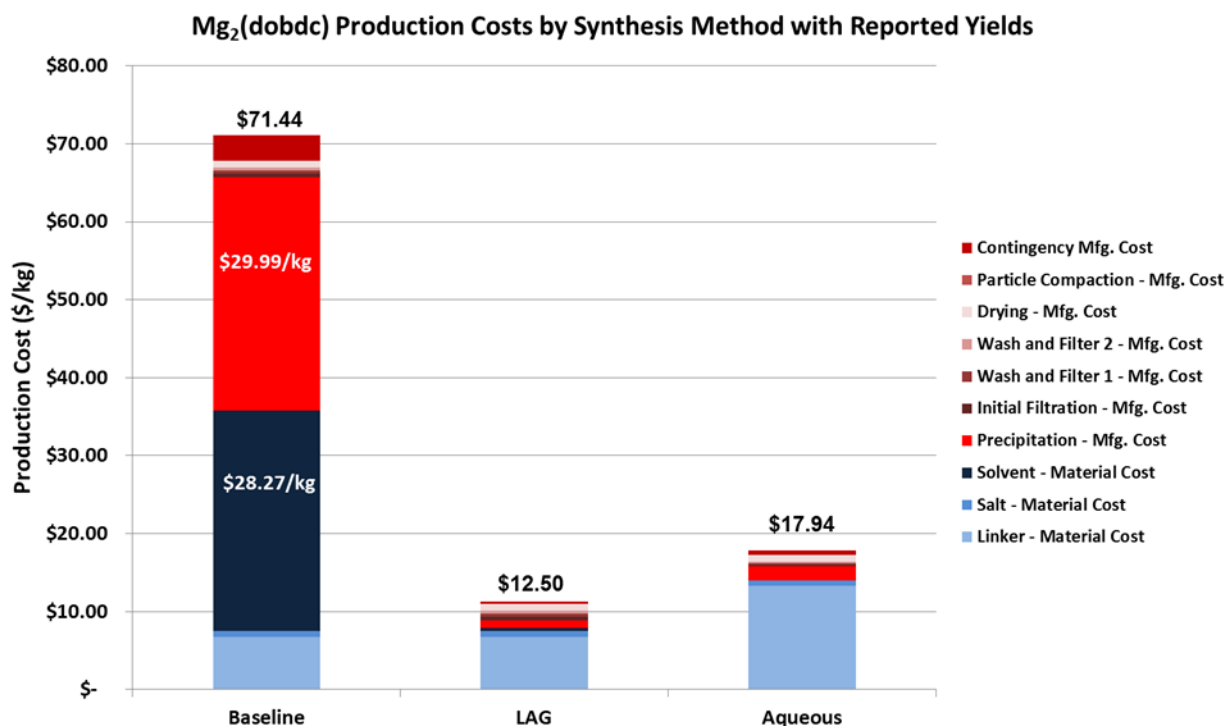


Figure 3 - Detailed cost breakdowns for the synthesis of Mg₂(dobdc) by the baseline industrial scale solvothermal, LAG, and aqueous synthesis methods. Material costs are represented by shades of blue while manufacturing costs are represented by shades of red. Yields are 69%, 69%, and 92% for the baseline, LAG, and aqueous methods, respectively.

Figure 3 shows cost breakdown comparisons for the three industrial scale synthesis methods analyzed, as applied to Mg₂(dobdc). This material serves as a representative MOF to illustrate cost trends. Yields for the baseline method and LAG are taken from lab yields reported by Tranchemontagne *et al.*, while aqueous yields are reported by Cadot *et al.*^{17,15} Costs are

divided between materials and manufacturing costs, and are further segregated by the processing steps shown in Figure 1. Linker costs do not vary between the baseline and LAG methods, except where the yield varies. Aqueous synthesis has a slightly higher linker cost than the LAG or baseline methods, owing to the variation in the stoichiometric requirements of

linker to salt listed in the laboratory procedures (see Table 8 in Supplemental Information). As a result, linker costs contribute \$6.74/kg for both the baseline and LAG synthesis, while the linker costs for aqueous synthesis is \$13.28/kg. The linker cost contribution in the aqueous synthesis methods could be reduced by up to 33% if the linker-to-metal ratio was reduced to the stoichiometric theoretical limit of 2 linkers per metal atom. The solvent and manufacturing costs are dramatically different between the baseline, LAG, and aqueous methods. Total organic solvent usage is reduced from approximately 95 wt% to 7.5wt% in LAG, resulting in a reduction of \$27.87/kg MOF for solvent cost. Organic solvents are eliminated altogether in the aqueous synthesis, resulting in a cost reduction of \$28.27/kg MOF for solvent costs. Manufacturing costs are also lower for LAG and aqueous synthesis methods. LAG methods reduce the cost of MOF manufacturing through a reduced reactor volume and reduced operation times. The combined reductions in volume and operating times reduce the number of simultaneous production lines required, resulting in a manufacturing cost reduction of \$31.92/kg. Aqueous manufacturing methods similarly reduce process operation time, which also reduces the number of simultaneous production lines required, resulting in a manufacturing cost reduction of \$31.52/kg.

thesis routes at 92% yield, so as to highlight fundamental differences, while eliminating yield based differences, due to different levels of MOF process optimization. Figure 4 shows cost breakdowns for materials and manufacturing compared for the three industrial scale synthesis methods at 92% yield. Both the baseline and LAG costs are lowered due to increasing the yield from 69% to 92%. Material costs (primarily linker and metal salt) are lowered by increased yield, but gains are modest due to the high reactant recycle rates assumed in the baseline process. Manufacturing costs are reduced more significantly by increased yield, owing to the lower capital expenditures for a given production rate.

Sensitivity Analysis

Sensitivity studies were conducted to further examine cost drivers beyond simple materials and synthesis methods, and to identify areas for future research that will further lower costs. Five parameters (shown in Figure 5), in addition to LAG and/or aqueous methods, were varied to determine which factors may have the largest impact on cost reduction. Switching processing from solvothermal to LAG or aqueous syntheses had the largest impact, as both approaches significantly reduce solvent cost. The next most influential parameter to the baseline cost is the mass ratio of salt-and-linker to solvent. This ratio, set at a baseline value of 2.5%, leads to high solvent usage and high reactor volume, and ideally would be much higher in a practical, compact, and cost efficient future plant.

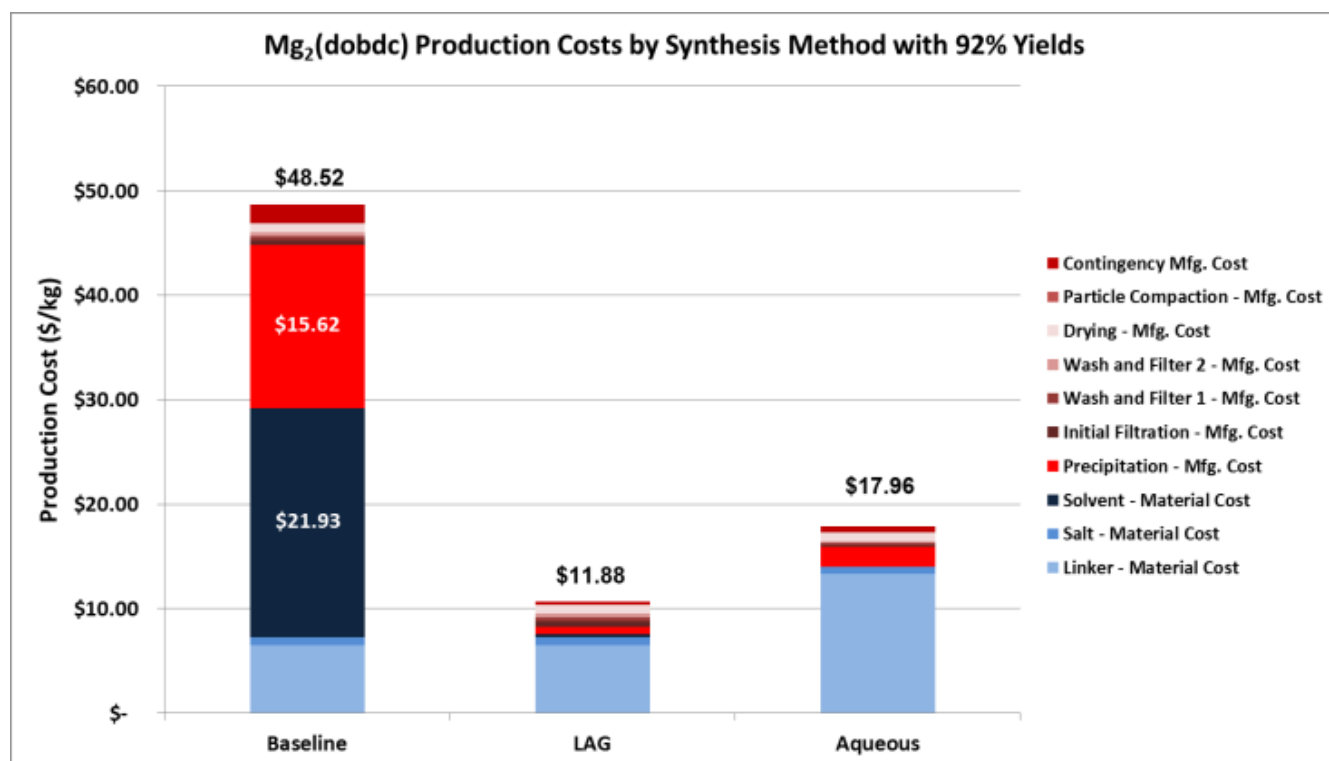


Figure 4 - Detailed cost breakdowns for synthesis of $\text{Mg}_2(\text{dobdc})$ by the baseline industrial scale solvothermal, LAG and aqueous synthesis methods at 92% yield. Material costs are represented by shades of blue while manufacturing costs are represented by shades of red. Aqueous linker costs are higher than LAG and baseline methods due to the higher salt:linker ratios used in the aqueous method.

Cost Comparison at Common Yield

Yields above 90% have been demonstrated by Das et al. and it is reasonable to expect all future MOFs reaching that approximate level when fully optimized.²⁸ Consequently, the MOF cost analysis was repeated with all four MOFs and three syn-

However, the ratio is limited by the inherent solubility of salt and linker in the solvent. Further research is required to explore solubility limits and to possibly find future low cost solvents with greater solubility characteristics. Predictably, yield is also a large cost driver, and any cost effective production

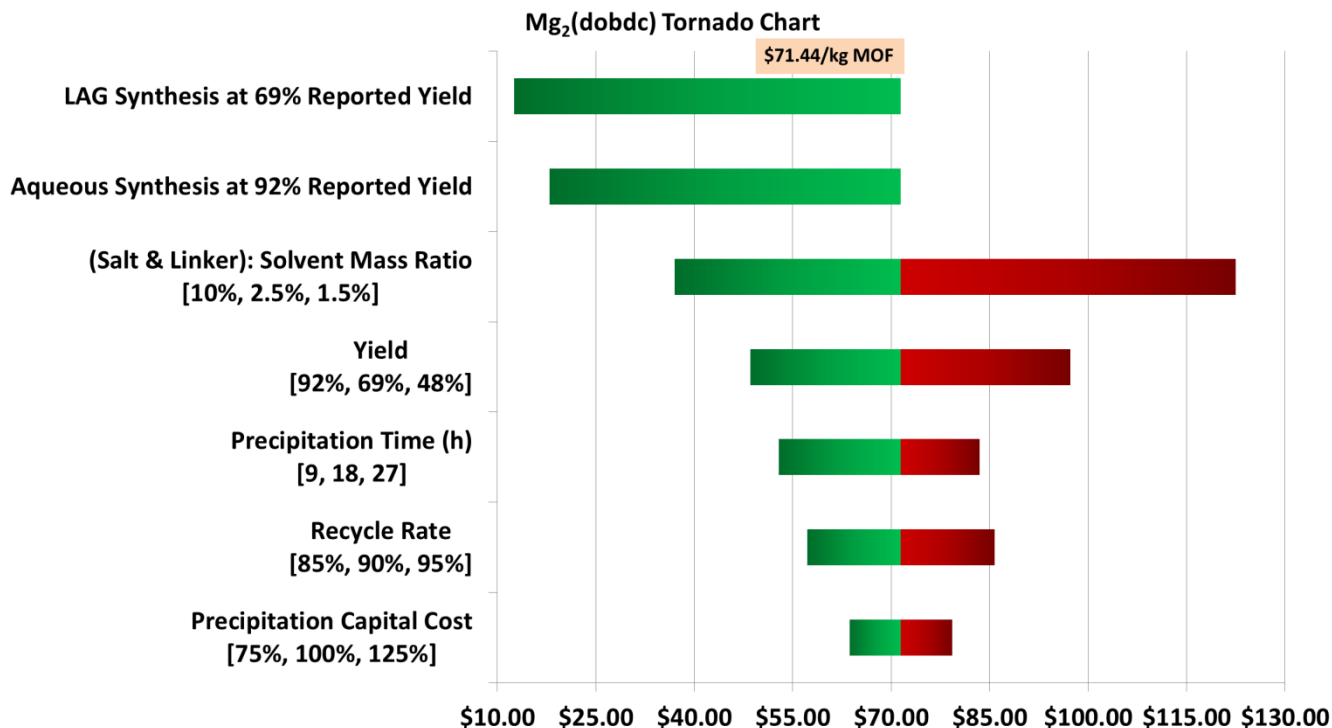


Figure 5 - Tornado chart showing parameter sensitivity for Mg₂(dobdc). Costs are taken at reported yields so as to identify the most significant factors including the need for optimization.

site will seek to maximize yield. By “stacking” and adjusting certain parameters from the sensitivity (as shown in Figure 6), a pathway to reducing the price of the MOF can be identified. Both LAG and aqueous routes, in combination with other improvements, hold promise to approach or be better than the \$10/kg MOF target. It is worth noting that adoption of the LAG or aqueous pathways alone are almost enough to reduce the cost of producing MOF below \$10/kg. In fact, the LAG method reduces the cycle time and the organic solvent usage so much that increasing the yield by 23% has little effect on the overall production cost.

Discussion

Solvent costs (even while assuming a solvent recycle rate of 90%) are a driving cost parameter of the industrial solvothermal baseline production method and account for more than 79% of the material costs for Mg₂(dobdc), 60% for Ni₂(dobdc), 69% for MOF-5, and 40% for HKUST-1. Achievement of high solvent recycle rates is, therefore, crucial, and the 90% recycle rate assumed for all materials (solvent, unreacted salt, and unreacted linker) in this analysis has not been publically demonstrated. Lowering the recycle rate would naturally raise the cost of production, as is further illustrated in Figure 5. Further research is needed to determine if such high recycle rates of organic solvents are feasible without affecting product quality.

Not surprisingly, reducing solvent usage will also be necessary in order to produce large quantities of MOF at low costs. LAG and aqueous methods lead to significantly lower materials and manufacturing costs compared to solvothermal methods. At 92% yield, Mg₂(dobdc) produced by LAG shows a 74% decrease in material costs (with 99% of the decrease being directly attributed to the solvent cost contribution) and an 84% decrease in manufacturing costs when compared to the base-

line solvothermal synthesis method. These steps alone almost achieve the ARPA-E target price of \$10/kg MOF, and only require an additional \$1.88/kg savings to make the target price.

The aqueous solution chemistry for Mg₂(dobdc) yields large material cost savings (up to 52%) and leads to a large reduction in total cost (up to 63%). Additional cost reduction opportunities exist through temperature and processing time optimization. The aqueous synthesis procedure requires the synthesis reactor contents to be held at reflux for 1 h. Maintaining reflux for such a large volume consumes a large amount of energy. As such, large and expensive heaters are required for the aqueous synthesis route, though the capital cost is less of an impact than the energy requirements. Optimization of the manufacturing process steps is a viable cost reduction strategy. Reducing the reflux time to 30 min would reduce the MOF cost by only ~2%. However, increasing the salt-and-linker:solvent ratio (currently at less than 3% for the baseline synthesis of Mg₂(dobdc)) to 10% offers a price reduction of approximately 50% in the baseline synthesis method.

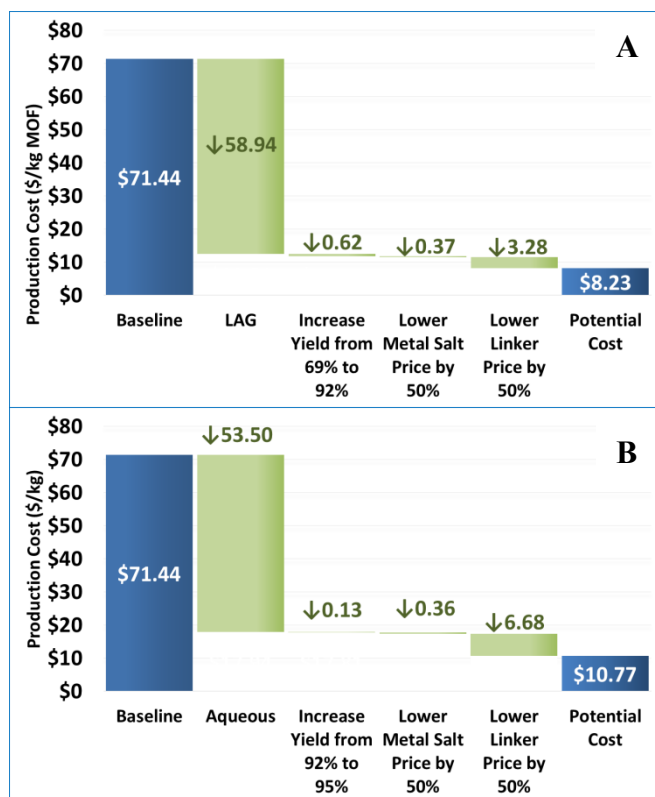


Figure 6 - Waterfall charts for $\text{Mg}_2(\text{dobdc})$ showing the potential costs of utilizing A) LAG and B) aqueous methods with reduced material prices to lower MOF costs.

Conclusions

The results of this study identify cost drivers and highlight two potential pathways to significantly reduced MOF production cost at an industrially relevant scale. Four MOFs were examined with generally similar cost trends thereby suggesting widespread applicability of the findings. Furthermore, these results quantify the impact of two synthesis alternatives (LAG and aqueous) to the traditional solvothermal approach. Implementation of LAG or aqueous synthesis is projected to lead to greater than 50% MOF cost reduction. When combined with other improvements, the projected costs approach the \$10/kg ARPA-E target for natural gas storage. These low cost MOFs would be invaluable in helping to achieve the DOE targets for a low cost H_2 storage system, as well as for many other potential MOF applications.

Both LAG and aqueous methods offer significant cost reductions, achieved primarily by reducing the use of organic solvent and, subsequently, manufacturing costs. Both methods are projected to achieve similar cost reductions, suggesting that both LAG and aqueous solution chemistries merit further process evaluation. Ultimately, the selection of synthesis method will be determined by which method fits best with a particular MOF. The possibility of combining LAG and aqueous methods into a single production method is worth further investigation.

It is also recognized that a fully continuous synthesis operation has the potential for further cost opportunities. For direct comparison and extension from lab scale, the scope of this analysis was based on a large-scale batch synthesis with certain steps that could be implemented with a pseudo-continuous operation such as drying and shaping. A future study should

be conducted to evaluate each process step to determine the most suitable approach between continuous and batch processing since certain batch process operations may still be optimal.

As solvent costs are a significant cost contributor, demonstration of high solvent recycle rates ($\geq 90\%$) is crucial to achieving even the moderate to high cost projections made within the analysis for solvothermal syntheses. This will be particularly important for MOFs that may not be amenable to aqueous or mechanochemical syntheses. Studies to minimize solvent usage are also recommended to both reduce material cost and reduce the size of reactors (and thereby reduce manufacturing cost).

Table 4- Comparison of MOF production prices for 50,000 systems per year at reported yields. Potential Cost values show the lowest potential price as described in the water fall charts above (See Figure 6).

		Thermosolvent	LAG	Aqueous	Potential Cost (LAG)	Potential Cost (Aqueous)
Total Cost	Ni ₂ (dobdc)	\$58.69	\$20.55	\$19.41	\$12.11	\$12.58
	Mg ₂ (dobdc)	\$71.44	\$12.50	\$17.94	\$8.23	\$10.77
	MOF-5	\$34.76	\$13.85	Not Analyzed	\$8.34	Not Analyzed
	HKUST-1	\$53.75	\$35.73	Not Analyzed	\$17.62	Not Analyzed
Material Cost	Ni ₂ (dobdc)	\$38.28	\$17.24	\$13.66	\$9.05	\$6.83
	Mg ₂ (dobdc)	\$36.11	\$9.09	\$14.13	\$5.10	\$7.05
	MOF-5	\$32.09	\$11.20	Not Analyzed	\$5.95	Not Analyzed
	HKUST-1	\$38.67	\$32.93	Not Analyzed	\$15.38	Not Analyzed
Manufacturing Cost	Ni ₂ (dobdc)	\$20.41	\$3.31	\$5.75	\$3.06	\$5.75
	Mg ₂ (dobdc)	\$35.33	\$3.41	\$3.81	\$3.15	\$3.71
	MOF-5	\$2.67	\$2.65	Not Analyzed	\$2.39	Not Analyzed
	HKUST-1	\$15.08	\$2.80	Not Analyzed	\$2.24	Not Analyzed

Finally, after solvents, linker costs are a large contributor to MOF costs. Overall process optimization may further lower the total MOF cost by improving the process yield. In addition, for Mg₂(dobdc) and Ni₂(dobdc), we note that analogous MOFs recently synthesized using a less-expensive isomer of the dobdc⁴⁻ linker, specifically 4,6-dioxido-1,3-benzenedicarboxylate, can be expected to significantly reduce MOF cost, while also boosting the H₂ and possibly NG storage capacities.⁷ The production methods described here have used fairly low temperatures (20 °C to 120 °C), with little consideration for pressure optimization, and only one examination of synthesis reaction time. Each of these parameters can have a significant effect on yield and (perhaps redundantly) cycle time. Cost estimates were largely projected for all MOFs with a 92% yield based on the assumption that process optimization could achieve that performance level, although this may not have been demonstrated for all MOFs.

More than 20,000 different MOF compounds have been reported in the literature and that number continues to grow. Existing and newly identified materials should have their costs and manufacturing processes critically assessed so as to identify cost drivers. Such assessments can then lead to process optimization, higher yields, lower material costs, and simplified synthesis. In addition, increases in MOF adsorption performance can also reduce the material quantity required within the system and lead to lower overall cost. Pursuing the recommendations outlined within this techno-economic analysis provides a pathway for researchers to achieve DOE and ARPA-E cost targets for MOFs in the future. The results of this analysis are expected to be generally valid for other MOFs and highlight avenues of investigation for the R&D community to address MOF costs.

AUTHOR INFORMATION

Corresponding Author

Daniel DeSantis
4075 Wilson Blvd
Suite 200
Arlington, VA 22203
ddesantis@sainc.com

Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Funding Sources

Natural gas storage research was funded by the Advanced Research Projects Agency – Energy (ARPA-E), U.S. Department of Energy, while hydrogen storage research in Berkeley was funded by the Department of Energy, Office of Energy Efficiency and Renewable Energy, Fuel Cell Technologies Office (under grant DE-AC02-05CH11231).

Disclaimer

While this article is believed to contain correct information, Ford Motor Company (Ford), and all authors, do not expressly or impliedly warrant, nor assume any responsibility, for the accuracy, completeness, or usefulness of any information, apparatus, product, or process disclosed, nor represent that its use would not infringe the rights of third parties. Reference to any commercial product or process does not constitute its endorsement. This article does not provide financial, safety, medical, consumer product, or public policy advice or recommendation. Readers should independently replicate all experiments, calculations, and results. The views and opinions expressed are of the authors and do not necessarily reflect those of Ford. This disclaimer may not be removed, altered, superseded or modified without prior author permission.

ABBREVIATIONS

MOF, Metal Organic Framework; NG, Natural Gas; H₂, Hydrogen, DOE, Department of Energy; AFV, Alternative Fuel Vehicles; LAG, Liquid Assisted Grinding; EtOH, Ethanol; DMF, Dimethylformamide; MeOH, Methanol

REFERENCES

1. Ordaz, G., Houchins, C. & Hua, T. *DOE Hydrogen and Fuel Cells Program Record*. Hydrogen.Energy.Gov (2015). at <http://hydrogen.energy.gov/pdfs/9014_hydrogen_storage_materials.pdf>
2. DOE. *Technical Assessment: Cryo-Compressed Hydrogen Storage for Vehicular Applications*. (2006). at <http://www.hydrogen.energy.gov/pdfs/cryocomp_report.pdf>
3. Wong, J. *Compressed hydrogen infrastructure*. (2005). at <http://ieahia.org/pdfs/Case-Studies/Compressed_Hydrogen_Infrastructure.aspx>
4. DOE. FCTO Multi-Year Research, Development, and Demonstration Plan. *Multi-Year Research, Development, and Demonstration Plan* (2013). at <<http://energy.gov/eere/fuelcells/downloads/fuel-cell-technologies-office-multi-year-research-development-and-22>>
5. ARPA-E. *MOVE Program Overview*. (2012). at <http://arpa-e.energy.gov/sites/default/files/documents/files/MOVE_ProgramOverview.pdf>
6. Viditha, V., Rao, M., Srilantha, K., Himabindu, V. & Yerramilli, A. A study on metal organic framework (MOF-177) synthesis, characterization and hydrogen adsorption-desorption cycles. *Int. J. Energy Environ.* **4**, 127–132 (2013).
7. Kapelewski, M. T. *et al.* M2(m-dobdc) (M = Mg, Mn, Fe, Co, Ni) Metal-Organic Frameworks Exhibiting Increased Charge Density and Enhanced H₂ Binding at the Open Metal Sites. *J. Am. Chem. Soc.* **2**, 12119–29 (2014).
8. Mason, J. A., Veenstra, M. & Long, J. R. Evaluating metal-organic frameworks for natural gas storage. *Chem. Sci.* **5**, 32–51 (2014).
9. Hydrogen Fuel Cell Vehicles Center for Climate and Energy Solutions. at <<http://www.c2es.org/technology/factsheet/HydrogenFuelCellVehicles>>
10. Collins, D. J. & Zhou, H.-C. Hydrogen storage in metal-organic frameworks. *J. Mater. Chem.* **17**, 3154 (2007).
11. Farha, O. *et al.* Metal-Organic Framework Materials with Ultrahigh Surface Areas: Is the Sky the Limit? *J. Am. Chem. Soc.* **134**, 15016–15021 (2012).
12. Service, R. F. Stepping on the Gas. *Science Magazine* 538–541 (2014). doi:10.1126/science.346.6209.538
13. Friščić, T. New opportunities for materials synthesis using mechanochemistry. *J. Mater. Chem.* **20**, 7599 (2010).
14. Friščić, T. *et al.* Ion- and Liquid-Assisted Grinding: Improved Mechanochemical Synthesis of Metal-Organic Frameworks Reveals Salt Inclusion and Anion Templating. *Angew. Chemie* **122**, 724–727 (2010).
15. Cadot, S., Veyre, L., Luneau, D., Farrusseng, D. & Alessandra Quadrelli, E. A water-based and high space-time yield synthetic route to MOF Ni₂(dhtp) and its linker 2,5-dihydroxyterephthalic acid. *J. Mater. Chem. A* **2**, 17757–17763 (2014).
16. Yilmaz, B., Trukhan, N. & Müller, U. Industrial Outlook on Zeolites and Metal Organic Frameworks. *Chinese J. Catal.* **33**, 3–10 (2012).
17. Tranchemontagne, D. J., Hunt, J. R. & Yaghi, O. M. Room temperature synthesis of metal-organic frameworks: MOF-5, MOF-74, MOF-177, MOF-199, and IRMOF-0. *Tetrahedron* **64**, 8553–8557 (2008).
18. Steward, D., Ramsden, T. & Zuboy, J. *H₂A Central Hydrogen Production Model, Version 3 User Guide*. National Renewable Energy Laboratory (2012). at <https://www.hydrogen.energy.gov/h2a_production.html>
19. Caskey, S. R., Wong-Foy, A. G. & Matzger, A. J. Dramatic tuning of carbon dioxide uptake via metal substitution in a coordination polymer with cylindrical pores. *J. Am. Chem. Soc.* **130**, 10870–10871 (2008).
20. James, B. & DeSantis, D. Personal Correspondance with Dr. Steven Kay of Mosaic Materials. (2015).
21. Matoga, D. New MOF - type layered coordination polymers of manganese, method of their preparation, modification and use thereof. 1–8 (2015).
22. James, S., McNally, T. & Haydon, R. Process for the preparation of a metal-organic compound. 1–19 (2014).
23. Friščić, T. *et al.* Real-time and in situ monitoring of mechanochemical milling reactions. *Nat. Chem.* **5**, 66–73 (2013).
24. Klimakow, M., Klobes, P., Thünemann, A. F., Rademann, K. & Emmerling, F. Mechanochemical synthesis of metal-organic frameworks: A fast and facile approach toward quantitative yields and high specific surface areas. *Chem. Mater.* **22**, 5216–5221 (2010).
25. Friščić, T., Halasz, I., Štrukil, V., Eckert-Maksić, M. & Dinnebier, R. E. Clean and Efficient Synthesis Using Mechanochemistry: Coordination Polymers, Metal-Organic Frameworks and Metallodrugs. *Croat. Chem. Acta* **85**, 367–378 (2012).
26. Garzón-Tovar, L., Carné-Sánchez, A., Carbonell, C., Imaz, I. & Maspoch, D. Optimised room temperature, water-based synthesis of CPO-27-M metal-organic frameworks with high space-time yields. *J. Mater. Chem. A* (2015). doi:10.1039/C5TA04923G
27. Doetze Jakob Sikkema, O. & Adrianus Maria Reichwein, A. Process for Dicarboxylating Dihydric Phenols. 0–3 (2000).
28. Das, A. K. *et al.* An Efficient Synthesis Strategy for Metal-Organic Frameworks: Dry-Gel Synthesis of MOF-74 Framework with High Yield and Improved Performance. *Sci. Rep.* **6**, 28050 (2016).

